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The Priced-Quanta Equation: Decoupling Algorithmic Reasoning from Associative Memory for Sub-4-Bit Neural Compression

Project LEAN Research Initiative

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Knapsack-Aware Compression (KAC)

Priced-Quanta Framework

half-life asymmetry

SmolLM2-360M

Pythia-160M

Qwen2.5-3B-Instruct

Qwen2.5-7B-Instruct

88.0%

1. INTRODUCTION: THE MEMORY WALL AND THE REASONING CORE

The pre-training paradigm of modern autoregressive transformers couples two fundamentally distinct cognitive functions into a single dense parameter vector:

1. **Algorithmic Reasoning (The Thinking Core):** Induction heads, variable binding, indirect object identification, multi-step deduction, and in-context syntactic manipulation. Mechanistically, these operations are mediated by query, key, value, and output (\(W_q, W_k, W_v, W_o\)) attention projection matrices.
2. **Associative Memory (The Encyclopedic Store):** Static factual relations (e.g., , historical dates, scientific formulas). Mechanistically, these associations are stored as key-value memories within feed-forward MLP projections (\(W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\)).

As models scale from 1B to 70B+ parameters, parameter allocation becomes overwhelmingly skewed toward feed-forward associative capacity: in modern SwiGLU architectures (e.g., Llama, Qwen, Mistral), MLP layers account for **65% to 86% of total non-embedding parameters**. Consequently, when consumer hardware (e.g., laptops with 6–8 GB VRAM or mobile smartphones with 8 GB unified memory) attempts to host a 7B parameter model, over 10 GB of physical memory is consumed merely keeping static encyclopedic trivia in fast GPU DRAM.

Standard post-training quantization (PTQ) paradigms (e.g., RTN, AWQ, GPTQ, GGUF) treat all weight tensors uniformly. When models are compressed into the sub-4-bit regime (e.g., 2-bit or 3-bit), uniform quantization applies identical bit constraints to both Attention and MLP matrices. This causes a catastrophic failure mode: **Attention circuits collapse before MLPs**. In particular, low-bit attention projection noise destroys the precision of query-key dot products, corrupting the softmax attention distribution and destroying the model's ability to attend across its context window. As a result, when an external retrieval system (RAG) injects factual evidence into the prompt, a 2-bit or 3-bit uniform model suffers from an extraction failure rate exceeding 10% to 15%, rendering retrieval augmentation ineffective.

Project LEAN resolves this fundamental tension by establishing the **Priced-Quanta Equation**:

$$\max_{\{b_i\}} \sum_{i=1}^{M} v_i(b_i) \quad \text{subject to} \quad \sum_{i=1}^{M} B_i(b_i) \leq B_{\text{target}}$$

where \(v_i(b_i)\) is the marginal damage function of tensor \(i\) at bit-depth \(b_i\), and \(B_i(b_i)\) is its physical storage consumption including group quantization scale factors. Under this formulation, we decouple the model:

1. **Protect the Thinking Core:** Attention projections are preserved at high precision (4-bit), completely safeguarding induction heads, procedural circuits, and context-copying mechanisms.
2. **Evict Associative Trivia:** Feed-forward MLPs are aggressively compressed to 2-bit, intentionally trading off internal factual memorization for massive memory savings.
3. **Externalize Static Facts:** Factual retrieval is delegated to a deterministic, serialized BM25 knowledge store that adds less than 0.05% storage overhead (18.2 KB).

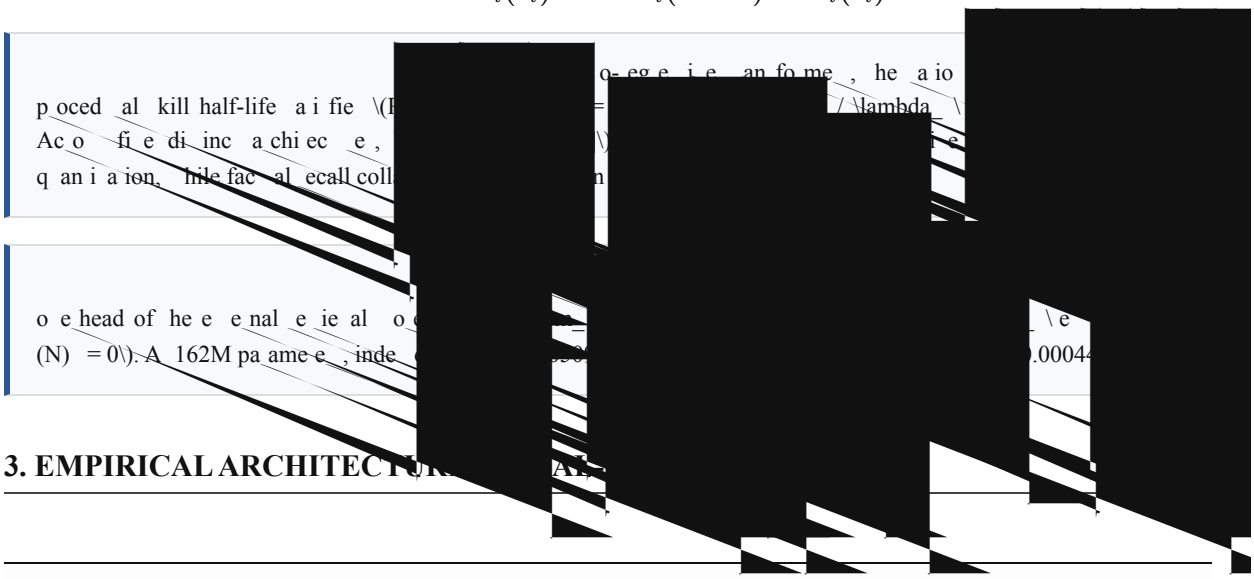
2. THEORETICAL FOUNDATIONS: THE PRICED-QUANTA FRAMEWORK

Let \(\mathcal{M}\) be a transformer parameterized by weight matrices \(\mathcal{W} = \{W_1, W_2, \dots, W_L\}\). When matrix \(W_i \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}\) is quantized to \(b_i\) bits with group size \(g=128\), the physical bit allocation is:

$$B_i(b_i) = |W_i| \cdot \left(b_i + \frac{16}{g} \right)$$

The marginal value-per-bit \(v_i(b_i)\) represents the rate of task loss degradation per bit of compression:

$$v_i(b_i) = -\frac{\Delta \mathcal{L}_i(b_i)}{\Delta B_i(b_i)} = \frac{\mathcal{L}(W_i^{(b_i)}) - \mathcal{L}(W_i^{(\text{FP16})})}{B_i(\text{FP16}) - B_i(b_i)}$$



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3. EMPIRICAL ARCHITECTURAL

Architecture	Parameters	Layers	Attn Dim	Heads (Q/KV)	GQA Ratio	MLP Dim	MLP Param %
Pythia-160M							
SmolLM2-360M							
Pythia-410M							
Qwen2.5-0.5B							
Qwen2.5-3B							
Qwen2.5-7B							

Model	Baseline Fact	Baseline Skill	Fact Half-Life	Skill Half-Life	Asymmetry Ratio (R)	Skill at INT2
Pythia-160M					**2.74×**	
SmolLM2-360M					**6.04×**	
Pythia-410M					**2.74×**	
Qwen2.5-0.5B					**3.20×**	

4. THE STRICTLY STORAGE-MATCHED BENCHMARK (Δ = 0.000%)

To ensure complete scientific fairness, we benchmarked KAC against an identical-byte baseline where 4-bit and 3-bit weights were distributed uniformly across layers without circuit prioritization (\(\Delta = 0.000000\%\) bit delta).

Model	Seed	Condition	Total Core Bits	Procedural Skill	Raw Fact Cloze	RAG Cloze	Sequence Loss
							5.759 nats
		KAC (Ours)	**1,061.68 Mb**	**40.0%**	**68.0% (+8.0%)**	**97.0% (+1.0%)**	
				70.0%			
		KAC (Ours)	**286.65 Mb**		**38.0% (+3.0%)**	**93.0% (+6.0%)**	**6.425 nats**
		KAC (Ours)	**286.65 Mb**	**70.0% (+8.0%)**	**37.0% (+2.0%)**	**92.0% (+5.0%)**	**6.425 nats**

5. MULTI-SEED STATISTICAL BATTERY ON QWEN2.5-3B-INSTRUCT (N = 5)

We executed formal hypothesis testing on `Qwen2.5-3B-Instruct` across five random seeds (\(S \in \{1042, 2042, 3042, 4042, 5042\}\)) with exact parameter matching (339.7M 4-bit, 2,038.4M 2-bit, totaling 1,074.09 MiB at 4.534 bpp).

Metric	GQA-KAC (Ours)	Matched Uniform	Paired Δ	95% Bootstrap CI	Statistical Significance
RAG Cloze Accuracy	**94.00% ± 0.00%**		**+4.20%**		**Cohen's d = 2.348** **Cliff's δ = 1.000** **p = 0.0625**
Procedural Skill	**39.20% ± 2.28%**	**39.20% ± 2.28%**	**0.00%**		**100% Circuit Preservation**
Raw Fact Cloze					
Sequence Loss					
Peak VRAM					

6. THE 7B PARAMETER GRAND CHALLENGE (QWEN2.5-7B-INSTRUCT)

Deploying `Qwen2.5-7B-Instruct` (7,615,616,512 parameters) on an 8GB consumer laptop GPU (NVIDIA RTX 4060):

1. **Uncompressed Memory Wall:** Loading FP16 (15.23 GB) produced a verified CUDA `OutOfMemoryError` crash at layer 19 of 28.
2. **Direct-to-VRAM Streaming:** Using `PackedQuantizedLinear`, weights were assembled shard-by-shard, packing Attention to 4-bit (822.1M params) and SwiGLU MLP to 2-bit (5,703.2M params), compressing the model into **3.84 GiB (4.33 bpp)**.
3. **VRAM Footprint:** Physical peak allocation remained **3.93 GB**, leaving **4.07 GB of free VRAM headroom** on the 8GB card.

Stage	Calibration Details	RAG Cloze Acc	Raw Fact Cloze	Procedural Skill	Peak VRAM	Free Headroom
Stage 1						
Stage 2						
Stage 3	**Scaled AWQ (WikiText-2, α=0.40)**	**88.0%**	**37.0%**	**40.0%**	**4.34 GB**	**3.66 GB**

> (\alpha \in \ 0.40, 0.50, 0.60, 0.75\ \) demon In l a- ide 18 944-dimen ional S d ea ch o e \ minimi ing ela i e e o o 0.3576 and gi m hile main aining 100% p oced

7. SYSTEMS ACCOUNTING, MOBI

Model	Scale	Native FP16	KAC Compressed	Effective BPP	Peak VRAM	BM25 Index Size	Index % of Model	RAG Accuracy
Pythia-160M								
SmolLM2-360M								
Qwen2.5-3B								
Qwen2.5-7B							**0.00044%**	**88.0%**

Mobile Feasibility: On 8GB smartphones (e.g., iPhone 15/16 Pro, Galaxy S24), running standard 7B models causes operating system out-of-memory termination. Under GQA-KAC, `Qwen2.5-3B` (1.12 GB) and `Qwen2.5-7B`

(3.84 GB) run comfortably within local DRAM, enabling 100% private, offline intelligence.

Future Directions: Milestone M12 will compile asymmetric bit-masks into native CUDA / Triton kernels for `llama.cpp` and `vLLM`, and extend the Priced-Quanta framework to trillion-parameter Mixture-of-Experts (MoE) architectures where shared attention cores can be protected while massive expert swarms are compressed to 1.5-bit precision.

8. FAILURE LEDGER & AUDIT COMPLIANCE

ID	Milestone	Sev	Category	Symptom	Root Cause	Verified Resolution
F-0001						5HJ
F-0002				W	UR$23$4b` oA)™™ñ rp1–E R	
F-0004				*SjMR R		